

Descriptor-Teacher Guided Multi-to-Uni Transfer for Low-Resource Regression ADMET Prediction

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Abstract

Low-resource ADMET prediction is a persistent challenge in early-stage drug discovery: labeled assays are scarce, while inexpensive molecular knowledge such as physicochemical descriptors is readily available. We study a training-time descriptor transfer setting in which descriptors may guide model learning but are not required by the deployed predictor. To this end, we develop Descriptor-Teacher AdapterFusion, a lightweight ECFP4-only student that combines a pseudo-descriptor adapter, descriptor reconstruction supervision, and prediction distillation from an ECFP4+descriptor random forest teacher. Across five regression ADMET datasets, three low-resource training ratios, and three random seeds, fixed teacher distillation with $\lambda_{\text{distill}} = 1.0$ improves AdapterFusion in 11/15 dataset-ratio settings, with an average RMSE change of -0.0656 relative to the undistilled AdapterFusion baseline. Descriptor-access random forests remain stronger in most settings, so the contribution is controlled descriptor knowledge transfer into an ECFP-only student rather than an overall ADMET state-of-the-art claim. A lambda sweep and adaptive-weighting ablation further show that naive validation-based adaptive distillation is not sufficient.

1 Introduction

Accurate ADMET prediction is important for early drug discovery because absorption, distribution, metabolism, excretion, and toxicity properties affect which candidate molecules can progress beyond screening [2, 18, 20]. Many ADMET tasks are low-resource: labels are expensive to obtain, measurements vary across assays, and each property may have a limited number of reliable annotated compounds. This motivates models that can use auxiliary molecular knowledge without complicating the predictor that is ultimately deployed.

Molecular descriptors provide one such source of auxiliary knowledge. RDKit physicochemical descriptors are inexpensive to compute and often correlate with ADMET behavior [13]. A direct way to exploit them is to concatenate descriptors with molecular fingerprints or use descriptor-based traditional models. This is an important descriptor-access setting, but it is not the only useful formulation. In applications where the deployed model should remain lightweight, descriptor-free, or compatible with an existing fingerprint-only pipeline, it is preferable to use descriptors as training-time knowledge while preserving ECFP-only inference. This creates a multi-to-uni transfer problem: can descriptor knowledge improve a model that ultimately uses only ECFP4 fingerprints [14]?

This paper studies descriptor-guided multi-to-uni transfer for low-resource regression ADMET prediction. We begin with an ECFP4 MLP baseline, compare simple auxiliary descriptor prediction with a structured AdapterFusion student, and then add teacher prediction distillation from an ECFP4+descriptor random forest. The central question is deliberately narrow: can descriptor-access knowledge improve an ECFP-only student under low-resource regression ADMET splits?

The results are positive but bounded. Teacher distillation improves AdapterFusion in most regression settings, while descriptor-access random forest controls remain strong.

The contributions are:

- We formulate low-resource regression ADMET prediction as a controlled training-time descriptor transfer problem with ECFP-only inference.
- We introduce Descriptor-Teacher AdapterFusion, which augments an ECFP-only AdapterFusion student with descriptor reconstruction and descriptor-access random forest teacher distillation.
- We evaluate descriptor prediction, AdapterFusion, descriptor-access controls, tree-based baselines, and teacher-distilled AdapterFusion under shared dataset-ratio-seed splits.
- We show that fixed descriptor-teacher distillation improves AdapterFusion in 11/15 regression dataset-ratio settings and report lambda sensitivity, validation-selected lambda analysis, and a negative adaptive-weighting ablation.

2 Related Work

Molecular representation learning has increasingly used auxiliary objectives to inject chemical knowledge into learned representations. Molecular language models such as MolBERT motivate domain-relevant pretraining and auxiliary tasks rather than relying only on generic representation learning objectives [5]. Graph pretraining methods such as GNN pretraining, GROVER, and GraphMVP similarly use self-supervised or cross-view objectives to improve downstream molecular prediction [8, 12, 15]. Our study does not introduce a large pretrained encoder. Instead, it isolates a narrower question: whether a cheap and interpretable descriptor modality can supervise an ECFP-only predictor under low-resource ADMET splits.

Auxiliary learning and task-specific adaptation are also relevant. A simple auxiliary objective may not transfer all relevant information to the downstream task; the architecture through which auxiliary information is routed can matter [4, 16]. The closest conceptual motivation is multi-to-uni molecular representation learning, including M2UMol-style knowledge transfer [21]. Our setting is controlled: descriptors are the auxiliary modality, ECFP4 is the inference modality, and ADMET prediction is the downstream task. We further add teacher-student distillation [7], using a descriptor-access molecular property model as the teacher rather than a larger neural encoder.

ADMET benchmark modeling provides the application context. We use TDC-style ADMET datasets and evaluate low-resource splits across regression and classification properties [9]. MoleculeNet provides an earlier benchmarking reference for molecular property prediction metrics, splits, and baselines [19]. Fingerprint and descriptor models remain necessary controls in cheminformatics, while graph neural networks and directed message-passing models provide stronger learned-representation baselines in larger-scale studies [6, 11, 22, 23]. Tree ensembles are included because random forests and gradient boosting methods remain competitive for tabular molecular features [1, 3, 10, 17].

3 Method

3.1 Problem setting

Each molecule has an ECFP4 fingerprint x_{ecfp} , an optional descriptor vector x_{desc} , and a target y . The inference constraint is that the student model uses only x_{ecfp} at test time. Descriptor

information may be used during training as auxiliary supervision or through teacher predictions. We focus on regression tasks, where the task loss is mean squared error. We use “descriptor-access” for models that consume true descriptors at inference time, and “ECFP-only” for students that do not.

3.2 ECFP encoder and descriptor prediction

The base ECFP model maps a 2048-dimensional ECFP4 fingerprint to a 128-dimensional latent representation using a two-layer MLP:

$$z_{\text{fp}} = f_{\text{fp}}(x_{\text{ecfp}}). \quad (1)$$

The simple descriptor-prediction baseline adds an auxiliary head that predicts standardized RDKit descriptors from z_{fp} . This baseline tests whether descriptor supervision alone is enough to shape the ECFP representation. Its training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\text{transfer}} \mathcal{L}_{\text{desc}}, \quad (2)$$

where $\lambda_{\text{transfer}} = 0.1$. The descriptor vector contains nine RDKit descriptors: MolWt, LogP, TPSA, HBA, HBD, rotatable bonds, aromatic ring count, heavy atom count, and ring count.

3.3 AdapterFusion student

The AdapterFusion student uses the same ECFP encoder but introduces a pseudo-descriptor adapter. Instead of forcing the downstream predictor to use only the original fingerprint representation, the student learns an intermediate representation intended to carry descriptor-like information:

$$z_{\text{desc}} = f_{\text{adapter}}(z_{\text{fp}}). \quad (3)$$

A learned fusion gate combines the original ECFP representation and the pseudo-descriptor representation:

$$g = \sigma(f_{\text{gate}}([z_{\text{fp}}, z_{\text{desc}}])), \quad (4)$$

$$z_{\text{fused}} = g \odot z_{\text{fp}} + (1 - g) \odot z_{\text{desc}}. \quad (5)$$

The target prediction head receives z_{fused} , while the descriptor prediction head predicts descriptors from z_{desc} . This design separates the source of descriptor supervision from the inference interface: the student is trained to form descriptor-informed features, but no true descriptors are consumed at test time.

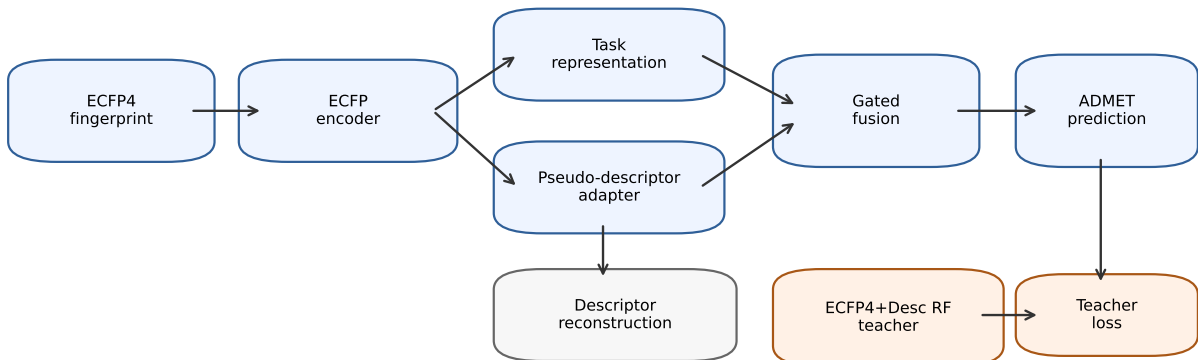
3.4 Descriptor-teacher distillation

We include descriptor-access controls to estimate the value of direct descriptor information. The strongest descriptor-access teacher used here is an ECFP4+descriptor random forest. This teacher is used only during training; the distilled student keeps the AdapterFusion architecture and adds a teacher prediction loss:

$$\mathcal{L}_{\text{teacher}} = \text{MSE}(\hat{y}_{\text{student}}, \hat{y}_{\text{teacher}}). \quad (6)$$

The final objective is:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\text{transfer}} \mathcal{L}_{\text{desc}} + \lambda_{\text{distill}} \mathcal{L}_{\text{teacher}}. \quad (7)$$



Training uses descriptor reconstruction and teacher prediction loss; inference uses only ECFP4.

Figure 1: Descriptor-Teacher AdapterFusion. Training uses descriptor reconstruction and teacher prediction distillation from an ECFP4+descriptor random forest. Inference uses only ECFP4.

The main configuration uses $\lambda_{\text{transfer}} = 0.1$ and $\lambda_{\text{distill}} = 1.0$, selected from the fixed lambda sweep over $\{0.01, 0.1, 0.3, 1.0\}$. The teacher here is specifically a descriptor-access molecular property model, so the distillation target encodes descriptor-informed prediction behavior while preserving ECFP-only inference for the student.

4 Experiments

4.1 Datasets and protocol

The main regression benchmark includes five TDC ADMET tasks: Caco-2 permeability, lipophilicity, aqueous solubility, volume of distribution, and plasma protein binding. Supplementary classification experiments cover five standard binary ADMET endpoints. The main claims are restricted to regression tasks because the descriptor-teacher distillation signal is more consistent in this setting.

Each dataset is evaluated with training ratios of 10%, 20%, and 50%. Each dataset-ratio setting is run with seeds 42, 123, and 3407. All models use the same dataset-ratio-seed splits, so comparisons are paired by construction. Descriptor scalers are fit only on the training split and then applied to validation and test splits.

4.2 Baselines and metrics

We compare ECFP4 MLP, ECFP4 MLP with descriptor prediction, AdapterFusion, descriptor-concatenation MLP, ECFP4 random forest, descriptor random forest, ECFP4+descriptor random forest, and Descriptor-Teacher AdapterFusion. Descriptor-concatenation and descriptor random forest models use descriptors at inference and are therefore descriptor-access controls rather than ECFP-only students. They provide a reference for how much predictive information is available when descriptors are directly accessible.

Regression tasks use MAE and RMSE, with RMSE as the primary metric. Classification tasks use ROC-AUC and PR-AUC and are reported only as supplementary reference results. All tables report mean $\pm$ standard deviation over three seeds.

Table 1: Main low-resource regression ADMET results. Values are test RMSE mean $\pm$ std over three seeds; lower is better. Δ is distilled AdapterFusion minus base AdapterFusion, so negative values indicate improvement.

Dataset	Train %	DescPred	AdapterFusion	ECFP4+Desc RF	Distilled AdapterFusion	Δ
caco2_wang	10	3.3584 $\pm$ 0.1667	1.7702 $\pm$ 0.0304	0.5535 $\pm$ 0.0017	1.7725 $\pm$ 0.0497	0.0023
caco2_wang	20	1.6383 $\pm$ 0.0859	0.9541 $\pm$ 0.0563	0.4691 $\pm$ 0.0083	0.9459 $\pm$ 0.0362	-0.0081
caco2_wang	50	0.7613 $\pm$ 0.0784	0.6775 $\pm$ 0.0995	0.4252 $\pm$ 0.0121	0.6265 $\pm$ 0.0508	-0.0510
lipophilicity_astrazeneca	10	1.2638 $\pm$ 0.0440	1.2325 $\pm$ 0.0114	1.0635 $\pm$ 0.0011	1.2058 $\pm$ 0.0219	-0.0267
lipophilicity_astrazeneca	20	1.1232 $\pm$ 0.0275	1.0960 $\pm$ 0.0132	0.9858 $\pm$ 0.0015	1.0903 $\pm$ 0.0083	-0.0057
lipophilicity_astrazeneca	50	1.0013 $\pm$ 0.0082	0.9736 $\pm$ 0.0059	0.9184 $\pm$ 0.0012	0.9605 $\pm$ 0.0057	-0.0130
solubility_aqsoldb	10	1.8330 $\pm$ 0.0212	1.7832 $\pm$ 0.0151	1.4996 $\pm$ 0.0059	1.7906 $\pm$ 0.0205	0.0074
solubility_aqsoldb	20	1.7247 $\pm$ 0.0477	1.7117 $\pm$ 0.0125	1.3951 $\pm$ 0.0021	1.7027 $\pm$ 0.0288	-0.0090
solubility_aqsoldb	50	1.6411 $\pm$ 0.0166	1.6107 $\pm$ 0.0088	1.3397 $\pm$ 0.0057	1.5963 $\pm$ 0.0087	-0.0144
vdss_lombardo	10	5.4896 $\pm$ 0.0889	5.4563 $\pm$ 0.0463	5.2699 $\pm$ 0.0051	5.4187 $\pm$ 0.0199	-0.0376
vdss_lombardo	20	5.5751 $\pm$ 0.1937	5.5086 $\pm$ 0.1260	5.2805 $\pm$ 0.0220	5.6795 $\pm$ 0.0473	0.1709
vdss_lombardo	50	5.3722 $\pm$ 0.0619	5.2838 $\pm$ 0.0352	5.0729 $\pm$ 0.0257	5.3103 $\pm$ 0.0309	0.0265
ppbr_az	10	65.1399 $\pm$ 20.3123	18.1281 $\pm$ 0.1596	14.2419 $\pm$ 0.0867	17.5888 $\pm$ 0.2565	-0.5393
ppbr_az	20	26.9134 $\pm$ 2.7840	16.6702 $\pm$ 0.3270	13.7317 $\pm$ 0.0288	16.4194 $\pm$ 1.1199	-0.2508
ppbr_az	50	15.3150 $\pm$ 0.3506	14.9058 $\pm$ 0.5019	13.3106 $\pm$ 0.0968	14.6711 $\pm$ 0.7282	-0.2347

Table 2: Fixed distillation lambda ablation over 15 regression dataset-ratio settings. Negative mean delta indicates improvement over base AdapterFusion.

$\lambda_{distill}$	Complete	Beats Adapter	Mean Δ
0.01	15/15	8/15	0.0075
0.1	15/15	8/15	-0.0221
0.3	15/15	10/15	-0.0046
1.0	15/15	11/15	-0.0656

5 Results

5.1 Main regression results

Table 1 reports the main low-resource regression results. The descriptor-access random forest is a strong control across the benchmark, indicating that the selected RDKit descriptors contain useful ADMET-relevant information. This establishes the motivation for transfer: descriptor knowledge is predictive, but directly using descriptors changes the inference setting.

Among ECFP-only neural models, AdapterFusion is stronger than simple descriptor prediction in all 15 regression dataset-ratio settings. This suggests that the transfer path matters: predicting descriptors as a side task is less effective than routing the fingerprint representation through a pseudo-descriptor adapter and fusing it into the task representation.

Teacher distillation further improves AdapterFusion. With $\lambda_{distill} = 1.0$, distilled AdapterFusion improves over base AdapterFusion in 11/15 regression settings, with a mean RMSE delta of -0.0656. The gains are most visible in settings where the descriptor-access teacher has a large advantage over the ECFP-only student, such as **ppbr_az**. The distilled student still trails the descriptor-access RF teacher in most settings. The appropriate interpretation is therefore partial transfer of descriptor-access knowledge into an ECFP-only student, not a replacement for descriptor-access baselines.

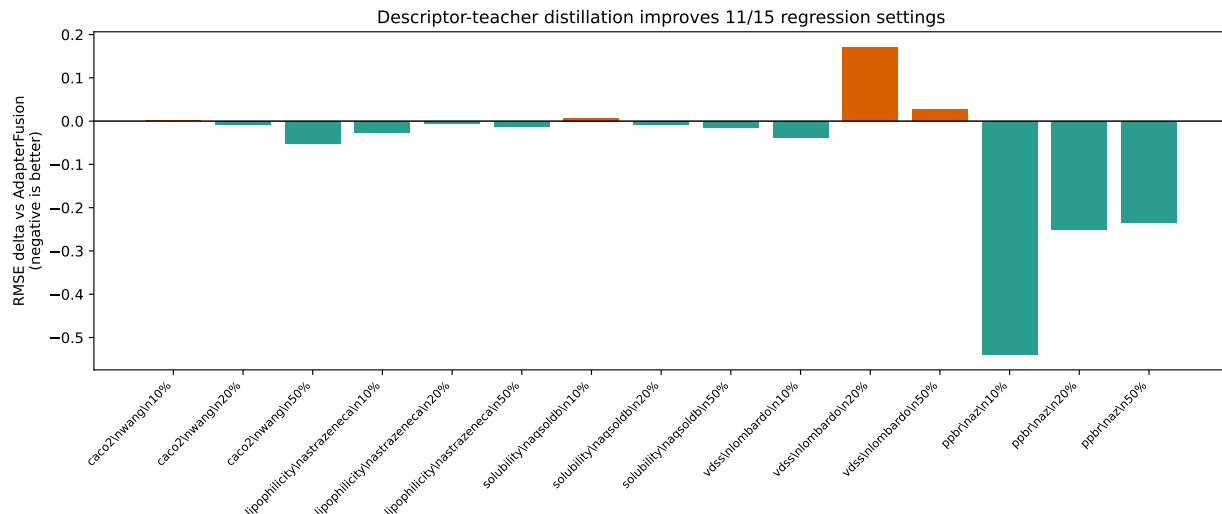


Figure 2: RMSE delta of distilled AdapterFusion relative to base AdapterFusion. Negative values indicate improvement. Fixed $\lambda_{\text{distill}} = 1.0$ improves 11/15 regression settings.

Table 3: Adaptive teacher-weighting negative ablation. The validation-advantage rule improves over base AdapterFusion in some settings, but is weaker than fixed strong distillation.

Comparison	Result
Adaptive beats base AdapterFusion	9/15
Adaptive beats fixed $\lambda = 1.0$	3/15
Adaptive beats best fixed lambda	1/15
Mean Δ vs fixed $\lambda = 1.0$	0.0817

5.2 Lambda sensitivity

Table 2 summarizes the fixed distillation lambda sweep. The strongest global fixed value is $\lambda_{\text{distill}} = 1.0$. Smaller lambdas can be better for individual dataset-ratio settings, but they do not improve aggregate performance. Validation-selected lambda is useful as a diagnostic analysis but beats fixed $\lambda_{\text{distill}} = 1.0$ in only 2/15 settings, while still beating base AdapterFusion in 12/15 settings. This supports using a simple fixed value as the main configuration and reporting lambda sensitivity as an ablation.

Figure 3 provides a diagnostic view of when distillation helps. A larger descriptor-teacher advantage can create room for improvement, but teacher advantage alone does not guarantee student gain. This motivates the adaptive-weighting analysis below and suggests that future variants should model teacher reliability more selectively.

5.3 Adaptive weighting as a negative ablation

Table 3 summarizes the validation teacher-advantage adaptive weighting rule. The adaptive rule improves over base AdapterFusion in 9/15 settings, but beats fixed $\lambda_{\text{distill}} = 1.0$ in only 3/15 settings and beats the best fixed lambda in only 1/15 settings. Its mean RMSE delta relative to fixed $\lambda_{\text{distill}} = 1.0$ is 0.0817, indicating worse average performance. This negative result is informative: a coarse validation-level teacher advantage is not a sufficient reliability signal for

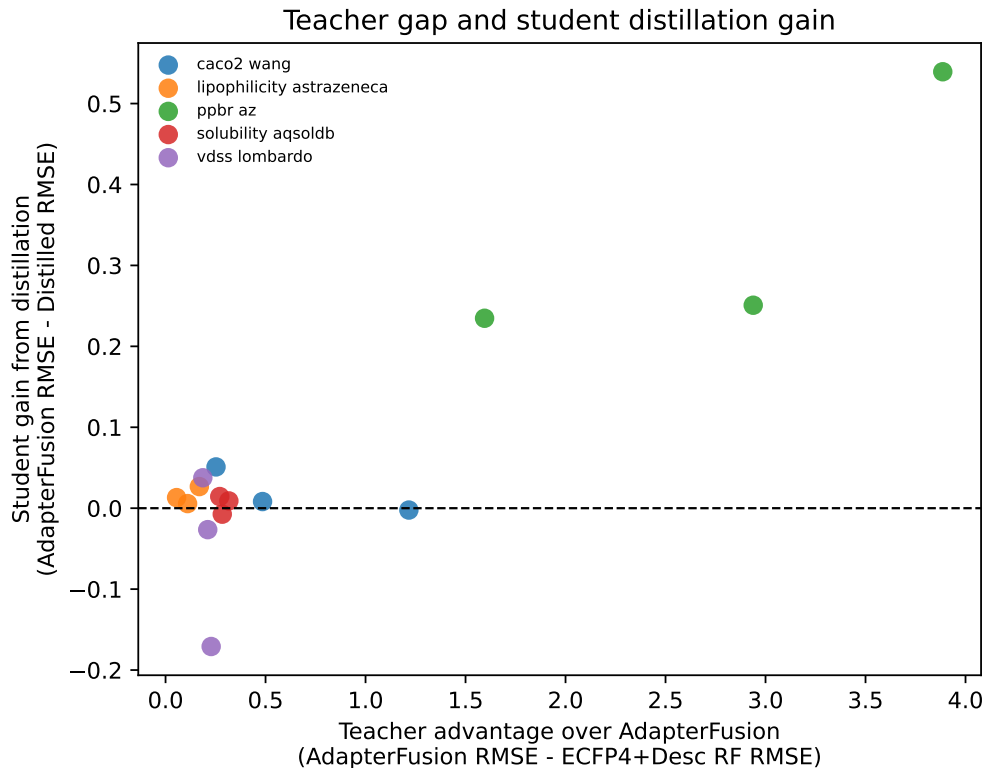


Figure 3: Relationship between descriptor-teacher advantage and student gain from distillation. Positive student gain means the distilled student has lower RMSE than base AdapterFusion.

adaptive distillation. More selective teacher weighting likely requires sample-level uncertainty, teacher-student disagreement, or learned reliability estimation.

5.4 Classification reference

Classification results are reported as supplementary reference results. They are less stable than the regression results, and descriptor-access controls are often strong. We therefore do not use classification results as the main evidence for Descriptor-Teacher AdapterFusion.

6 Discussion

The experiments support a controlled descriptor-guided transfer story. Descriptor information is valuable, and descriptor-access baselines remain strong. The contribution is not to replace descriptor-access random forests, but to show that descriptor-access knowledge can partially improve an ECFP-only neural student under low-resource regression ADMET settings.

The results also clarify which parts of the transfer mechanism matter. AdapterFusion improves over plain descriptor prediction, suggesting that the architecture used to route auxiliary knowledge is important. Teacher distillation provides an additional improvement by aligning the ECFP-only student with predictions from a descriptor-access random forest teacher. The lambda sweep shows that stronger teacher supervision is useful on aggregate, with $\lambda_{\text{distill}} = 1.0$ as the best fixed global setting. However, the dataset-specific best lambda varies, and neither validation-selected

lambda nor the tested adaptive rule clearly improves over the fixed setting. This points to a remaining methodological challenge: useful descriptor teachers should be trusted selectively, but simple validation-level rules are not reliable enough.

7 Limitations

This work has several limitations. First, the main claim is restricted to regression ADMET tasks. Classification results are included only as supplementary evidence because improvements are less consistent. Second, the student model is lightweight and ECFP-based; the method has not yet been tested with large pretrained molecular encoders. Third, descriptor-access random forests remain stronger than the distilled student in most settings, which limits the claim to transfer under ECFP-only inference rather than superiority over descriptor-access models. Finally, the adaptive teacher-weighting strategy tested here is intentionally simple and performs worse than fixed distillation; more principled sample-level or learned teacher weighting remains future work.

8 Conclusion

We presented Descriptor-Teacher AdapterFusion for low-resource regression ADMET prediction with ECFP-only inference. The method trains an AdapterFusion student with descriptor reconstruction and teacher prediction distillation from an ECFP4+descriptor random forest. With $\lambda_{\text{distill}} = 1.0$, the distilled student improves AdapterFusion in 11/15 regression dataset-ratio settings while still trailing descriptor-access random forests in most cases. These results support descriptor-teacher distillation as a useful but bounded mechanism for transferring descriptor knowledge into lightweight fingerprint-only models.

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